



ENGINEERING EXPLORATION FOR COMPUTER SCIENCE AND ENGINEERING

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WHAT IS ENGINEERING?

- Engineering is the application of science, mathematics, and creativity to design, build, and improve things that solve real-world problems.

ENGINEERING = SCIENCE + MATH + CREATIVITY + PRACTICAL PROBLEM-SOLVING

TYPES:

- Computer Science Engineering (CSE)
- Electronics & Communication Engineering (ECE)
- Electrical Engineering (EEE)
- Mechanical Engineering
- Civil Engineering
- Agricultural Engineering
- Instrumentation Engineering

ENGINEERING

- The application of scientific knowledge and mathematical principles to design, build, and optimize systems, structures, machines, and technologies that solve practical problems.
- Applies "why" to build something
- eg:using electron flow to build a circuit uses science to design & create
- Goal: practical, working solutions
- Method: design, build, test, improve

SCIENCE

- The systematic study of the natural world through observation, experimentation, and analysis to discover fundamental principles and laws.
- Discovers "why"
- e.g., why electrons flow to create current
Studies nature, facts & laws
- Goal: knowledge and understanding
- Method: observation, experiment, theory

SCIENTIST

- A scientist seeks to discover new knowledge by observing, experimenting, and analyzing natural phenomena. Their goal is to explain why things happen and uncover universal laws.
- Asks "why" — e.g., why sound patterns form speech
- Explores unknowns through research
- Works in labs, publishes findings
- Focus: discovering new knowledge

ENGINEER

- An engineer uses that knowledge to design and build practical solutions. Their goal is to solve problems, create systems, and make technology work reliably in the real world.
- Asks "how to build it" — e.g., how to build a device that understands speech
- Applies research to create products
- Works in industry, builds real system
- Focus: creating practical, working systems

NEEDS AND WANTS OF COMPUTER SCIENCE ENGINEER IN TODAY'S WORLD

- **GOOD CODING & PROBLEM-SOLVING SKILLS:**
Needed to write efficient programs and clear technical interviews
- **BASIC AI/MACHINE LEARNING KNOWLEDGE:**
Helps build smart features like chatbots and recommendation systems.
- **CLOUD & NETWORKING SKILLS:**
Most software today runs online and connects across devices.
- **CYBERSECURITY AWARENESS:**
Needed to protect user data and systems from attacks.
- **ABILITY TO WORK WITH OTHER ENGINEERING TEAMS:**
Hardware products like robots need CSE to collaborate with ECE/Mechanical.
- **WILLINGNESS TO KEEP LEARNING NEW TECH:**
Technology changes fast, so skills must be regularly updated.
- **GOOD COMMUNICATION SKILLS:**
Helps explain technical ideas clearly to non-technical teams.

VARIOUS DISCIPLINES OF ENGINEERING

- Computer Science Engineering (CSE) — designs software, algorithms, and systems that make devices intelligent
- Electronics & Communication (ECE) — builds circuits, sensors, and communication systems for signal processing
- Electrical Engineering (EEE) — designs power generation, distribution, and battery systems
- Mechanical Engineering — builds machines, structures, and moving mechanical parts
- Civil Engineering — designs and constructs buildings, roads, and infrastructure
- Agricultural Engineering — applies engineering to improve farming methods and equipment
- Instrumentation Engineering — designs sensors and measurement systems for accurate data collection

CASE STUDY: ROBOT

1. WHAT IS A ROBOT & WHY WE NEED IT?

- A robot is a machine that can sense its surroundings, process information, and act — often without constant human control.
- We need robots to handle tasks that are dangerous, repetitive, or beyond human reach — like exploring Mars, working in factories, or assisting in hospitals.
- Purpose: to extend what humans can safely and efficiently do, in places or situations we can't handle ourselves.

2. HOW A ROBOT CONNECTS WITH OTHER ENGINEERING BRANCHES

- CSE — writes the software/AI that lets the robot understand data and make decisions
- ECE — builds the sensors, cameras, and communication systems it uses to "see" and "hear"
- EEE — designs the power and battery systems that keep it running
- Mechanical — builds its body, wheels/legs, and moving parts
- Instrumentation — calibrates sensors so it accurately measures its surroundings
- No single branch can build a robot alone — every robot is the result of several branches working together.

NASA PERSEVERANCE ROVER

HOW PERSEVERANCE IS BUILT & WORKS

- A six-wheeled Mars rover, launched July 30, 2020, landed February 18, 2021. Powered by a nuclear battery (MMRTG) producing ~110 watts, plus two backup lithium-ion batteries. Runs on a radiation-hardened computer using flight software, and communicates via antennas relayed through Mars orbiters.

MAIN FUNCTIONS

- Searches for signs of ancient life
- Collects rock/soil samples for future return to Earth
- Tests oxygen production from Mars' atmosphere
- Uses 7 science instruments (cameras, X-ray tools, radar)
- Deployed the Ingenuity helicopter — first powered flight on another planet

Robotics in Everyday Life



Cleaning Robot
Assistant



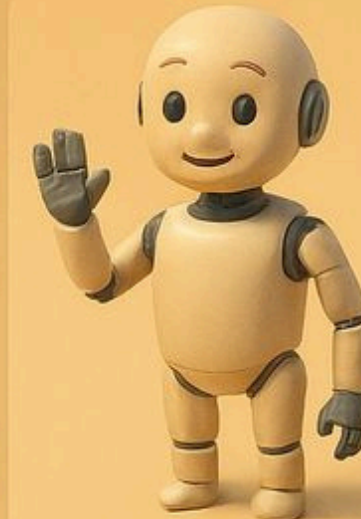
Kitchen
Cooking Robot



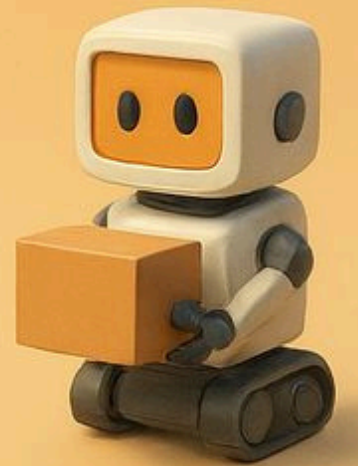
Delivery Drone
Robot



Window-Washing
Robot



Elderly Care
Humanoid



Warehouse
Sorting Robot



Gardening
Robot



Pet Companion
Robot



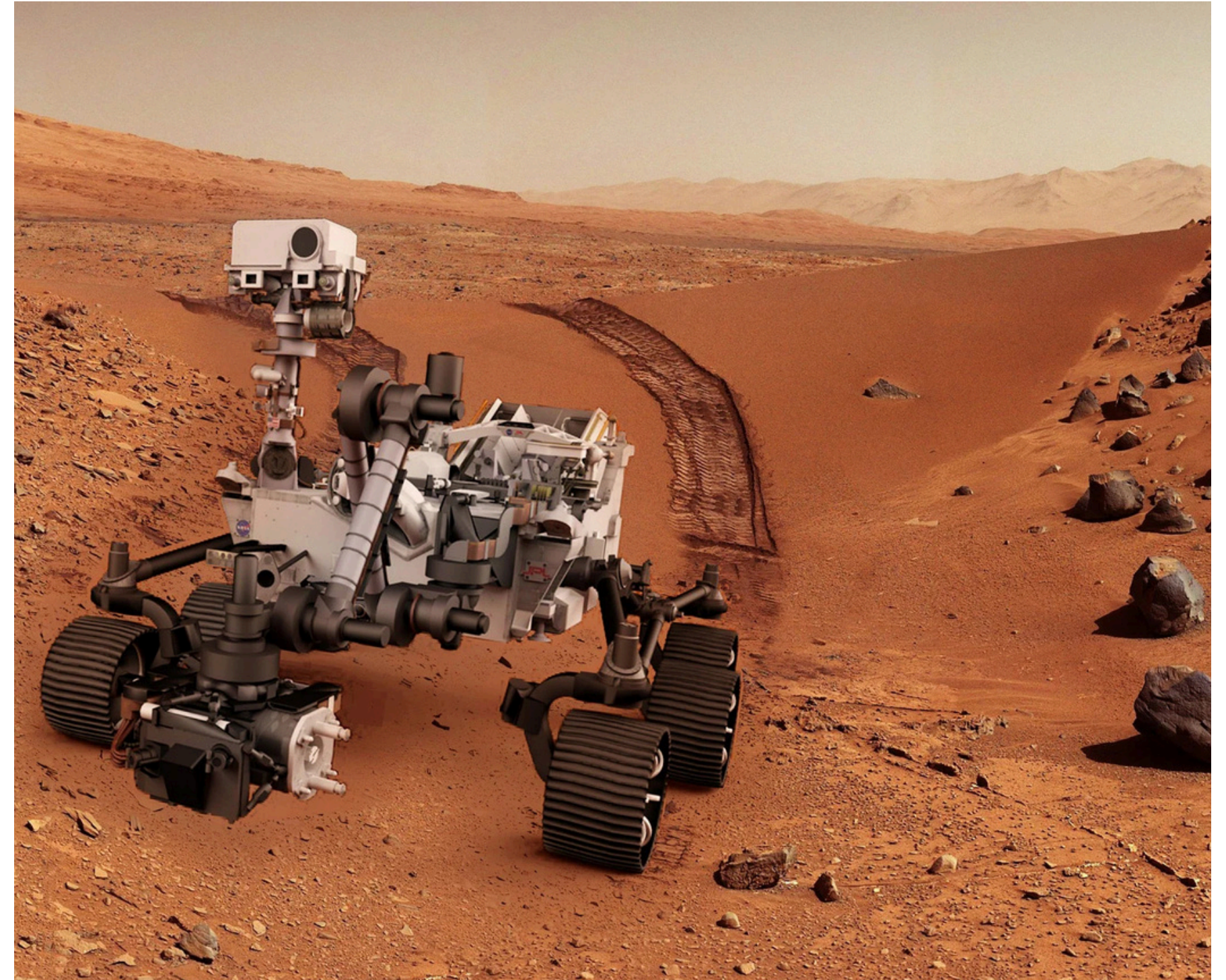
Construction Site
Robot

BRANCHES INVOLVED

- CSE — software, autonomous navigation, decision-making
- ECE — cameras, antennas, communication
- EEE — nuclear power system, battery management
- Mechanical — wheels, motors, robotic drill arm
- Instrumentation — instruments measuring rocks, weather, radiation

SIGNIFICANCE

- Perseverance must think, move, and survive fully on its own, since signals take ~11 minutes to reach Earth — proving no single branch could build it alone; only teamwork across disciplines makes it possible.



MISCONCEPTIONS ABOUT COMPUTER SCIENCE ENGINEER

- CSE is only about coding — it also covers AI, networks, security, and hardware interfacing.
 - CSE engineers don't need hardware knowledge — real products need them working with electronics/mechanical teams.
 - Anyone can self-learn CSE without a degree — strong DSA/OS/DBMS foundations are needed for complex problems.
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- CSE work is the same as being a scientist — CSE engineers build practical systems, not just research theories.
 - CSE engineers just sit and type code all day — the job also needs problem-solving, debugging, and teamwork.
 - A CSE degree guarantees a high-paying job automatically — skills and practical experience matter just as much.

SKILLS REQUIRED FOR COMPUTER SCIENCE ENGINEERING IN THE 21ST CENTURY:

- Strong DSA & problem-solving — core to writing efficient code and clearing interviews.
- AI/Machine Learning basics — powers robots, assistants, and recommendation systems.
- Cloud & networking skills — most software today runs online and connects devices.
- Cybersecurity awareness — protecting user data is now essential knowledge.
- Cross-branch collaboration — working with ECE/Mechanical teams on hardware products.
- Continuous learning ability — technology changes fast, so skills must keep updating.



THANK YOU!

